Diseases in RAG & Their Solutions

1. Hallucination

- **Problem**
    - LLM generates incorrect or fabricated information.
    - Retrieved documents do not support the answers.
- **Solutions**
    - Improve retrieval quality.
    - Use grounded prompting.
    - Generate citation-based answers.
    - Add confidence scoring.

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2. Poor Retrieval

- **Problem**
    - Relevant documents are not retrieved.
    - Irrelevant documents are returned.
- **Solutions**
    - Use better embedding models.
    - Apply Hybrid Search (Vector + Keyword).
    - Tune retrieval parameters.
    - Use document re-ranking.

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3. Context Pollution

- **Problem**
    - Retrieved context contains unnecessary information.
    - Model gets distracted by irrelevant content.
- **Solutions**
    - Retrieve fewer, high-quality chunks.
    - Remove duplicate chunks.
    - Apply a similarity threshold.
    - Use re-ranking.

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4. Lost in the Middle

- **Problem**
    - Important information in the middle of the context is ignored.
- **Solutions**
    - Place important chunks at the beginning.
    - Compress the context.
    - Reorder retrieved documents.
    - Use long-context LLMs.

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5. Chunking Problems

- **Problem**
    - Important information gets split across chunks.
    - Chunks are too large or too small.
- **Solutions**
    - Use semantic chunking.
    - Add chunk overlap.
    - Optimize chunk size.
    - Preserve document structure.

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6. Embedding Mismatch

- **Problem**
    - Query and document embeddings do not match semantically.
- **Solutions**
    - Use the same embedding model.
    - Use domain-specific embeddings.
    - Fine-tune embeddings if required.

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7. Vocabulary Gap

- **Problem**
    - User and documents use different terminology.
- **Solutions**

    - Query rewriting.
    - Query expansion.
    - Hybrid retrieval.
    - Synonym mapping.

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8. Outdated Knowledge

- **Problem**
    - Vector database contains stale information.
- **Solutions**
    - Periodic re-indexing.
    - Incremental updates.
    - Remove outdated embeddings.
    - Automate document refresh.

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9. Duplicate Retrieval

- **Problem**
    - Multiple retrieved chunks contain the same information.
- **Solutions**
    - Remove duplicate chunks.
    - Use Maximal Marginal Relevance (MMR).
    - Diversify retrieved results.

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10. Low Recall

- **Problem**
    - Relevant documents are missed.
- **Solutions**
    - Increase Top-K retrieval.
    - Improve embeddings.
    - Query expansion.
    - Hybrid search.

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11. Low Precision

- **Problem**
    - Too many irrelevant documents are retrieved.
- **Solutions**
    - Lower Top-K.
    - Apply metadata filtering.
    - Re-rank retrieved documents.
    - Tune similarity threshold.

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12. Context Window Limitation

- **Problem**
    - Retrieved context exceeds the model's input limit.
- **Solutions**
    - Context summarization.
    - Context compression.
    - Recursive retrieval.
    - Use long-context models.

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13. Multi-Hop Reasoning Failure

- **Problem**
    - Answer requires combining information from multiple documents.
- **Solutions**
    - Multi-step retrieval.
    - Graph RAG.
    - Agentic RAG.
    - Iterative retrieval.

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14. Ambiguous Queries

- **Problem**
    - User query is unclear or incomplete.
- **Solutions**

    - Ask clarification questions.
    - Query rewriting.
    - Intent detection.
    - Use conversation history.

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15. No Retrieved Context

- **Problem**
    - Retriever fails to find relevant documents.
- **Solutions**
    - Lower similarity threshold.
    - Expand the query.
    - Use keyword search as fallback.
    - Return "I don't know" when appropriate.

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16. Latency

- **Problem**
    - Slow retrieval and response generation.
- **Solutions**
    - Cache embeddings.
    - Cache responses.
    - Optimize vector database.
    - Parallelize pipeline components.

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17. Scalability Issues

- **Problem**
    - Performance decreases as document volume increases.
- **Solutions**
    - Distributed vector databases.
    - Efficient indexing (HNSW, IVF).
    - Database sharding.
    - Load balancing.

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18. Security & Data Leakage

- **Problem**
    - Unauthorized users access sensitive information.
- **Solutions**
    - Role-Based Access Control (RBAC).
    - Metadata-based permissions.
    - Encryption.
    - Audit logging.

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19. Prompt Injection

- **Problem**
    - Malicious documents manipulate the LLM's behavior.
- **Solutions**
    - Prompt sanitization.
    - Input/output filtering.
    - Ignore instructions from retrieved documents.
    - Use trusted data sources.

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20. Evaluation Difficulty

- **Problem**
    - Difficult to measure RAG performance accurately.
- **Solutions**
    - Use RAG evaluation metrics.
    - Human evaluation.
    - Benchmark datasets.
    - Frameworks like **RAGAS** for automated evaluation.